

Reg. No.

B.Tech. DEGREE EXAMINATION, MAY 2019

Third Semester

CS0201 – DATA STRUCTURES USING C++

(For the candidates admitted from the academic year 2007-2008 to 2012-2013)

Time: Three hours

Max. Marks: 100

Answer **ALL** Questions

PART – A (10 × 2 = 20 Marks)

1. Define ADT.
2. How do you measure the efficiency of the algorithm?
3. Define circular linked list.
4. Write a pseudocode for traversing an element in the doubly linked list.
5. Differentiate the conditions of stack size for overflow and underflow.
6. List down the applications of stack.
7. Differentiate between binary tree and complete binary tree.
8. Define graph with an example.
9. List down the phases of external sorting.
10. Write the worst case and average case order of complexity for quick sort.

PART – B (5 × 16 = 80 Marks)

11. a. Explain linear search and binary search with the help of pseudocode for given values 20,10,1,6,12,8,4,15.

(OR)

- b. Explain hashing and its operation with an example.

12. a. Write an algorithm for inserting and deleting an element from singly linked list.

(OR)

- b. Explain doubly linked list and its operations with examples.
13. a. Write the algorithm for converting infix expression to postfix expression.

(OR)

- b. What is queue? Explain its operation with example.

14. a. Construct a binary tree to satisfy the following orders:
inorder : D B F E A G C L J H K
Postorder : D F E B G L J K H C A

(OR)

- b. Describe the categories of AVL rotations.

15. a. Illustrate heap sort algorithm with an example.

(OR)

- b. Trace the algorithm of quick sort in step by step manner with an example.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2019
Third Semester

18CSC201J – DATA STRUCTURES AND ALGORITHMS
(For the candidates admitted during the academic year 2018-2019 onwards)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)

Answer ALL Questions

1. What is the advantage of bubble sort over other sorting techniques?
(A) It is faster (B) Consumes less memory
(C) Detects whether the input is already sorted (D) Consumes more memory
2. How many passes does an insertion sort algorithm consist of?
(A) N (B) N – 1
(C) N + 1 (D) N²
3. What is the worst case complexity of binary search using recursion?
(A) O (n log n) (B) O (log n)
(C) O (n) (D) O (n²)
4. Which of the following are themselves a collection of different data types?
(A) String (B) Structures
(C) Char (D) Arrays
5. Which of these best describes an array?
(A) A data structure that shows a hierarchical behaviour (B) Contains objects of similar types
(C) Contains objects of mixed types (D) Both similar and mixed types
6. A linear collection of data elements where the linear node is given by means of pointer is called
(A) Linked list (B) Node list
(C) Primitive list (D) Array list
7. What is the time complexity of searching for an element in a circular linked list?
(A) O (n) (B) O (n log n)
(C) O (1) (D) O (n²)
8. Which of the following is not a disadvantage to the usage of array?
(A) Fixed size (B) You know the size of the array prior to allocation
(C) Insertion based on position (D) Accessing elements at specified positions

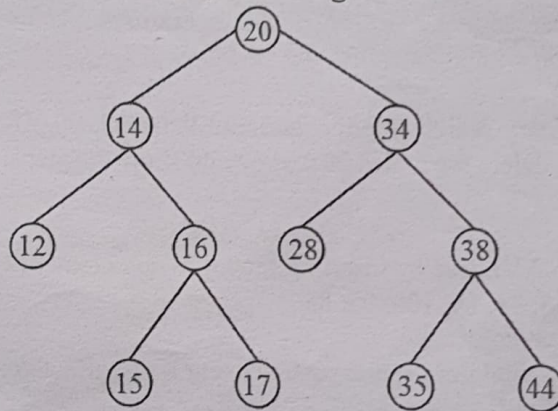
9. Process of inserting an element in a stack is called _____.
 (A) Create (B) Push
 (C) Evaluation (D) Pop
10. What is the time complexity of pop() operation when the stack is implemented using an array?
 (A) $O(1)$ (B) $O(n)$
 (C) $O(\log n)$ (D) $O(n \log n)$
11. What data structure would you most likely to see in a non-recursive implementation of a recursive algorithm?
 (A) Linked list (B) Stack
 (C) Queue (D) Tree
12. To implement a stack using queue with only enqueue and dequeue operations, how many queues will you need?
 (A) 1 (B) 2
 (C) 3 (D) 4
13. A node in a binary tree can have atmost _____ children.
 (A) 2 (B) Any number of children
 (C) 0 or 1 or 2 (D) 0 or 1
14. Given an empty AVL tree, how would you construct AVL tree when a set of numbers are given without performing any rotations?
 (A) Just build the tree with the given input (B) Find the median of the set of elements given, make it as root and construct the tree
 (C) Use trial and error (D) Use dynamic programming to build the tree
15. The tree traversal to obtain a prefix expression is _____.
 (A) Level-order traversal (B) Pre-order traversal
 (C) Post-order traversal (D) In-order traversal
16. What is the maximum height of an AVL tree with P nodes?
 (A) P (B) $\log(P)$
 (C) $\log(P) / 2$ (D) $P/2$
17. What is direct addressing?
 (A) Distinct array position for every possible key (B) Fewer array positions than keys
 (C) Fewer keys than array positions (D) A structure used for storage
18. A graph with all vertices having equal degree is known as _____.
 (A) Multi graph (B) Regular graph
 (C) Simple graph (D) Complete graph
19. What kind of deletion is implemented by hashing using open addressing?
 (A) Active deletion (B) Standard deletion
 (C) Lazy deletion (D) Passive deletion

20. A graph having an edge from each vertex to every other vertex is called _____.
(A) Tightly connected (B) Strongly connected
(C) Weakly connected (D) Loosely connected

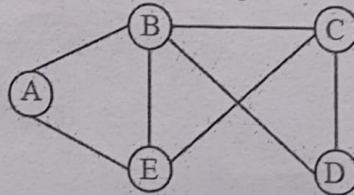
PART – B ($5 \times 4 = 20$ Marks)

Answer ANY FIVE Questions

21. Mention the best case, worst case and average case of bubble sort algorithm.
22. Write the pseudocode for linear search.
23. List the difference between arrays and linked lists.
24. Define Dequeue. How is it represented? What are the types of dequeue?
25. List any four applications of stacks.
26. Perform preorder and inorder traversal for the tree given below.



27. Consider the graph given below and find out the degree of each node.



PART – C ($5 \times 12 = 60$ Marks)

Answer ALL Questions

28. a. Write the pseudo code for insertion sort and apply the same to the following elements.
39, 9, 45, 63, 18, 81, 108, 54. Mention its best case, worst case and average case complexity.

(OR)

- b.i. Write the recursive pseudocode for binary search. (5 Marks)
ii. Explain the various mathematical notations and functions used in analysis of algorithms with proper graphs. (7 Marks)

29. a. Draw the node structure of a doubly linked list. Write the pseudocode for the following operations on a doubly linked list.

- (i) Delete the last node
- (ii) Insertion after a given node
- (iii) Insertion at the end

(OR)

b. Describe the following operations of circular linked list with an example

- (i) Insertion at the beginning
- (ii) Insertion after a given node
- (iii) Deletion at the end

30. a.i. Convert the given expression into postfix expression $9 - ((3 * 4) + 8) / 4$.

ii. Convert the given expression into prefix expression $(A - B / C) * (A / K - L)$.

(OR)

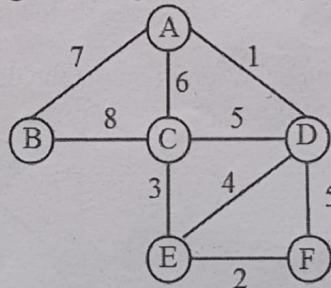
b. Elaborate the enqueue and dequeue operations performed in queue using linked list representation with algorithms.

31. a. Construct binary search tree for the input list: 100, 90, 45, 80, 30, 35, 20, 60, 70. Trace the algorithm to delete the nodes 20 and 60 and show the steps.

(OR)

b. Construct an AVL tree by inserting the following elements in order.
65, 3, 15, 28, 14, 100, 95, 85

32. a. Construct the minimum spanning tree using Kruskal's algorithm for the following graph.



(OR)

b.i. Consider a hash table with size = 10, using linear probing and insert the keys 72, 27, 36, 24, 63, 81, 92 and 100 in the table. (8 Marks)

ii. List the pros and cons of quadratic probing. (4 Marks)

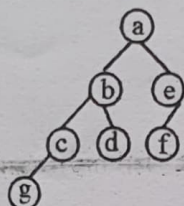
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8. Find the output of the following code snippet for the list $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$? 1 2 2 1
- ```
void solve (struct node* start)
{
 if (start == null) return;
 printf ("%d", start->data);
 if (start->next != null)
 {
 solve (start->next->next);
 }
 printf ("%d", start->data);
}
```
- (A) 1 2 3 4 5 6                      (B) 1 3 5 5 3 1  
(C) 1 3 5 1 3 5                      (D) 2 4 6 1 3 5
9. Consider the following code snippet 1   1   2   1
- ```
struct node
{
    int data; struct node * next;
}
typedef struct node NODE;
NODE * ptr;
```
- (A) $\text{ptr} = (\text{NODE}^*) \text{malloc} (\text{sizeof} (\text{NODE}));$ (B) $\text{ptr} = (\text{NODE}^*) \text{malloc} (\text{NODE});$
(C) $\text{ptr} = (\text{NODE}^*) \text{malloc} (\text{sizeof} (\text{NODE}));$ (D) $\text{ptr} = \text{malloc} (\text{NODE} (\text{NODE}^*));$
10. _____ is the linked list in which none of the nodes contain a null pointer? 1 1 2 1
- (A) Circular linked list (B) Singly linked list
(C) Doubly linked list (D) Priority list
11. Convert the infix expression to postfix expression $(A + B) / (C - D)$ 1 3 3 1
- (A) $AB + CD / -$ (B) $AB + / CD -$
(C) $AB / + CD -$ (D) $AB + CD - /$
12. The condition to represent circular queue as full is _____. 1 1 3 1
- (A) $\text{Front} = \text{max} - 1$ and $\text{rear} = \theta$ (B) $\text{Front} = \theta$ and $\text{rear} = \theta$
(C) $\text{Front} = \text{max} - 1$ and $\text{rear} = \text{max} - 1$ (D) $\text{Front} = \theta$ and $\text{rear} = \text{max} - 1$
13. If the elements "A", "B", "C", "D" and "E" are placed in a stack and are popped one at a time, in which order will they be popped? 1 3 3 1
- (A) A B C D E (B) A B D E C
(C) C A B E C (D) E D C B A
14. Choose the correct sequence of elements after following the sequence of operations performed on stack: push (100), push(10), push(20), pop, push(20), pop, pop, pop, push(20), pop 1 4 3 1
- (A) 20, 20, 10, 100, 20 (B) 20, 100, 10, 20, 20
(C) 100, 10, 20, 20, 20 (D) 20, 20, 20, 10, 100
15. Which of the following permutations can be obtained in the output (in the same order) using a stack assuming that the input is the sequence 1, 2, 3, 4, 5 in that order? 1 4 3 1
- (A) 5, 4, 3, 1, 2 (B) 1, 5, 2, 3, 4
(C) 3, 4, 5, 2, 1 (D) 3, 4, 5, 1, 2

16. A binary search tree contains the values 1, 2, 3, 4, 5, 6, 7, 8. The tree is traversed in post-order and the values are printed out. Which of the following sequence is a valid output? 1 2 4 1
- (A) 5, 3, 1, 2, 6, 4, 8, 7 (B) 1, 2, 3, 4, 5, 6, 7, 8
(C) 8, 7, 6, 5, 4, 3, 2, 1 (D) 5, 3, 2, 4, 1, 6, 7, 8

17. A binary search tree is generated by inserting in order of the following integers: 50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. Number of nodes in the left sub tree and right sub tree of the root respectively is? 1 3 4 1
- (A) (4, 7) (B) (7, 4)
(C) (3, 3) (D) (8, 3)

18. In the balanced binary tree in the figure given below, how many nodes will become unbalanced when a node is inserted as a child of the node "g"? 1 2 4 1



- (A) 1 (B) 3
(C) 7 (D) 10
19. A binary search tree contains the numbers 1, 2, 3, 4, 5, 6, 7, 8. When the tree is traversed in preorder and the values in each node is printed out, the sequence of values. Obtained is 5, 3, 1, 2, 4, 6, 8, 7. If the tree is traversed in post order, the sequence is? 1 3 4 1
- (A) 2, 1, 4, 3, 7, 8, 6, 5 (B) 2, 1, 4, 3, 6, 7, 8, 5
(C) 1, 2, 3, 4, 8, 7, 6, 5 (D) 7, 8, 6, 4, 2, 1, 3, 5

20. In a binary max heap containing 'N' numbers, the smallest element can be found in time 1 2 4 1
- (A) $\theta(n)$ (B) $\theta(n \log n)$
(C) $\theta(n \log \log n)$ (D) $\theta(1)$

21. Graphs are represented using _____. 1 1 5 1
- (A) Adjacency tree (B) Adjacency graph
(C) Adjacency queue (D) Adjacency linked list

22. A graph with all vertices having equal degree is known as _____. 1 1 5 1
- (A) Multi graph (B) Regular graph
(C) Simple graph (D) Complete graph

23. Maximum clique of size '2' is possible with _____ graph. 1 2 5 1
- (A) Perfect (B) Tree
(C) Cartesian (D) Histogram

24. The number of elements in adjacency matrix of a graph having 7 vertices is 1 2 5 1
- (A) 7 (B) 77
(C) 14 (D) 49

25. A vertex with one degree is called _____ vertex. 1 1 5 1
- (A) Pendant (B) Isolated
(C) Source (D) Simple

PART – B (5 × 10 = 50 Marks)

Answer ALL Questions

Marks BL CO PO

26. a. Develop a sequential search algorithm for unordered data. Illustrate with an example and mention its best, worst and average case time complexity.

10 4 1 1

(OR)

- b. Write an algorithm to sort given list in $O(n)$ time and justify the time complexity with suitable example.

10 4 1 1

3, 8, 5, 4, 1, 9, -2

27. a. Given two polynomials represented by a linked list $4x^3 + 7x^2 + 9$ and $15x^3 + 2x^2 + 4$, write a procedure that ADD two polynomials.

10 2 2 1

(OR)

- b. Discuss the following operations of doubly linked list with examples

10 2 2 1

- (i) Insert a node at middle
- (ii) Delete a value given by a user from list
- (iii) Modify a value in the list with user provided value

28. a.i. Write an algorithm to transform infix expression to prefix expression.

5 5 3 1

- ii. Consider the following arithmetic, infix expression Q:

5 5 3 1

$Q: A + (B * C - (D | E \uparrow F) * G) * H$ into equivalent prefix expression P.

(OR)

- b. Explain the insertion and deletion operation in circular queue with example.

10 5 3 1

29. a. Define binary search tree? Construct the same using the elements given
45, 39, 56, 12, 34, 78, 32, 10, 89, 81

10 1 4 1

Find in-order, pre-order and post-order traversal of the same tree.

(OR)

- b. Define AVL tree? Construct the same by inserting given elements in the same order.

10 1 4 1

18, 81, 29, 15, 19, 26, 25 and 1

Delete 15 after inserting all elements in the tree show the changes in the tree as. The insertion and deletion operation progresses.

30. a. A hash function 'b' defined by $h(key) = key \bmod 17$, with linear probing and quadratic probing to insert keys 44, 45, 76, 55, 91, 18, 63, 108 into a table. What will be the location of keys in the table? Illustrate.

10 3 5 1

(OR)

- b. Provide necessary algorithm to construct a minimum spanning tree using prim's logic for the given undirected weighted graph.

10 3 5 1

